

Auteur: Michael Livchits
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 43.4

$$\int \frac{2x+1}{(x^2-1)^2} dx$$

$$(x^2-1)^2 = (x-1)^2(x+1)^2$$

$$\frac{2x+1}{(x-1)^2(x+1)^2} = \frac{A}{(x-1)^2} + \frac{B}{(x+1)^2} + \frac{C}{x-1} + \frac{D}{x+1}$$

$$A = \frac{2x+1}{(x+1)^2} \text{ met } x = 1$$

$$A = \frac{3}{4}$$

$$B = \frac{2x+1}{(x-1)^2} \text{ met } x = -1$$

$$B = -\frac{1}{4}$$

Vervolgens om C te vinden

$$\begin{aligned} & \frac{2x+1}{(x-1)^2(x+1)^2} - \frac{A}{(x-1)^2} \\ &= \frac{8x+4}{4(x-1)^2(x+1)^2} - \frac{3(x+1)^2}{4(x-1)^2(x+1)^2} = \frac{-3x^2+2x+1}{4(x-1)^2(x+1)^2} \\ &= \frac{(x-1)(3x+1)}{4(x-1)^2(x+1)^2} = \frac{3x+1}{4(x-1)(x+1)^2} \end{aligned}$$

$$B = \frac{3x+1}{4(x+1)^2} \text{ met } x = 1$$

$$B = \frac{1}{4}$$

Vervolgens om D te vinden

$$\begin{aligned} & \frac{2x+1}{(x-1)^2(x+1)^2} - \frac{B}{(x+1)^2} \\ &= \frac{8x+4}{4(x-1)^2(x+1)^2} + \frac{(x-1)^2}{4(x-1)^2(x+1)^2} = \frac{x^2+6x+5}{4(x-1)^2(x+1)^2} \\ &= \frac{(x+1)(x+5)}{4(x-1)^2(x+1)^2} = \frac{x+5}{4(x-1)^2(x+1)} \end{aligned}$$

$$D = \frac{x+5}{4(x-1)^2} \text{ met } x = -1$$

$$D = \frac{1}{4}$$

Nu de integraal

$$I = \frac{3}{4} \int \frac{dx}{(x-1)^2} - \frac{1}{4} \int \frac{dx}{(x+1)^2} + \frac{1}{4} \int \frac{dx}{x-1} + \frac{1}{4} \int \frac{dx}{x+1}$$

Stel $u = x - 1$ en stel $t = x + 1$

$$du = dx \text{ en } dt = dx$$

$$\begin{aligned} I &= \frac{3}{4} \int \frac{du}{u^2} - \frac{1}{4} \int \frac{dt}{t^2} + \frac{1}{4} \ln|x-1| + \frac{1}{4} \ln|x+1| + C \\ &= -\frac{3}{4(x-1)} + \frac{1}{4(x+1)} + \frac{1}{4} \ln|x-1| + \frac{1}{4} \ln|x+1| + C \\ &= -\frac{x+2}{2(x^2+1)} + \frac{1}{4} \ln|x-1| + \frac{1}{4} \ln|x+1| + C \end{aligned}$$

References